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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/278,815	08/24/2023	Fook Yuen LEE	A4126.10056US01	5319

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EXAMINER	
RODGERS, THOMAS RAYMOND	

ART UNIT	PAPER NUMBER
3723	

NOTIFICATION DATE	DELIVERY MODE
05/27/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/278,815	Applicant(s) LEE, Fook Yuen	
	Examiner TOM R RODGERS	Art Unit 3723	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 5/4/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-15 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-15 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 8/24/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

The Examiner acknowledges the amendments. The previous rejections are withdrawn. New rejections are set forth herein and are made final.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 1, 2, 4, 10, and 12- 14 are rejected under 35 U.S.C. 103 as being unpatentable over Lorenz (US 4,310,377) in view of Rueb (US 5033797) in view of Morlock (EP3603446a1) in view of Polese (US 6,216,845).

Regarding claim 1 and 13, Lorenz discloses a brush manufacturing machine comprising: a rotary transportation mechanism (Item 6; a bristle carrier loading station; one or more bristle dispensing station (Item 2) comprising a bristle feed mechanism, a bristle tuft picker, and a bristle plant feeder; a fusion station; a heating station (Item 3); an unloading station (Item 5); and

a logic controller system for providing programmed commands, controlling a machine sequence (Column 6 Lines 32-57).

Lorenz fails to explicitly disclose one or more bristle dispensing station comprising a bristle feed mechanism, a bristle tuft picker, and a bristle plant feeder,; AND an ultrasonic fusion station; AND monitoring feedback of operating conditions at every station through a network of sensors and actuators

Rueb teaches a brush manufacturing machine wherein a bristle dispensing station comprising a bristle feed mechanism (Item 15), a bristle tuft picker (Item 17), and a bristle plant feeder (Items 10, 13, and 20); AND an ultrasonic fusion station (Column 19 Line 48-54).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Lorenz to adapt the bristle dispensing station of Rueb. Rueb discusses in Column 3 Lines 14-16 that this modification of preparing the bristles and gathering the bristles into an array is simple and economical.

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the type of welding/fusion to be an ultrasonic fusion station. When fixing the bristles to the main bristle head, there are many different known methods. These include, melting and curing, welding, or adhesive (or other known methods). Some methods have their own benefits in regards to the properties or design of the brush. Using a method of welding versus ultrasonic welding is beneficial when using any plastic sort of material, because ultrasonic welding can weld to areas beneath the surface to increase the bond between different components.

Morlock teaches a brush manufacturing machine wherein a logic controller system for providing programmed commands, controlling the machine sequence and monitoring feedback of operating conditions at every station through a network of sensors and actuators (Paragraph 23-53 discusses various scenarios in which sensors can be used). It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Lorenz to incorporate a variety of sensors as taught by Morlock. The use of various sensors is viewed as a manner to make a manual process automated, which has been held to be of

routine by one skilled in the art (see MPEP 2144.04). By understanding how the system is working, it would prevent unplanned down time (Morlock Paragraph 3).

Lorenz further fails to explicitly disclose wherein the bristle carrier loading station comprises a vibrating feeder bowl configured to receive bristle carriers, wherein the vibrating feeder bowl causes vibration provided by a rotary vibrating exciter, and wherein the vibration causes the bristle carriers to move in an upward spiral until the bristle carriers are oriented correctly and transported along a track to the bristle carrier loading station for pick-up. (Lorenz does discuss in Column 5 Lines 45-68 about a vibrational mechanism to help organize the magazine, but does not show or go into much further detail because these are known in the art).

Polese teaches a brush manufacturing machine wherein the bristle carrier loading station comprises a vibrating feeder bowl configured to receive bristle carriers, wherein the vibrating feeder bowl causes vibration provided by a rotary vibrating exciter, and wherein the vibration causes the bristle carriers to move in an upward spiral until the bristle carriers are oriented correctly and transported along a track to the bristle carrier loading station for pick-up (Item 20 Column 4 Lines 20-44). It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Lorenz to include the bowl feeder as taught by Polese. Vibration orientating machines are known in the art to allow an increase in productivity without the need of a workers constant presence (Polese Column 2 Lines 23-27).

Regarding claim 2, Lorenz in view of Rueb in view of Morlock in view of Polese disclose the brush manufacturing machine according to claim 1, wherein the bristle carrier loading station comprises a vibrating feeder bowl (Lorenz Column 5 Lines 52-61), an orientation check feature (Morlock Paragraph 42-45), a gripper arm (Rueb Item 18a) having a presence sensor that controls the movement of the gripper arm (Morlock Paragraph 42-45).

Regarding claim 4, Lorenz in view of Rueb in view of Morlock in view of Polese disclose the brush manufacturing machine according to claim 1, wherein the bristle tuft picker comprises

a picker plate with a picker face with one or more through holes with dimensions consistent with the through holes of the bristle carrier openings, the shape of the through holes is semi-circular (Rueb Figure 2 Item 18).

Regarding claim 10, Lorenz in view of Rueb in view of Morlock in view of Polese disclose the brush manufacturing machine according to claim 1, wherein the heating station comprises a heating element having a heating plate (Lorenz Item 51 and 52) shaped to match the bristle carrier profile and dimensions.

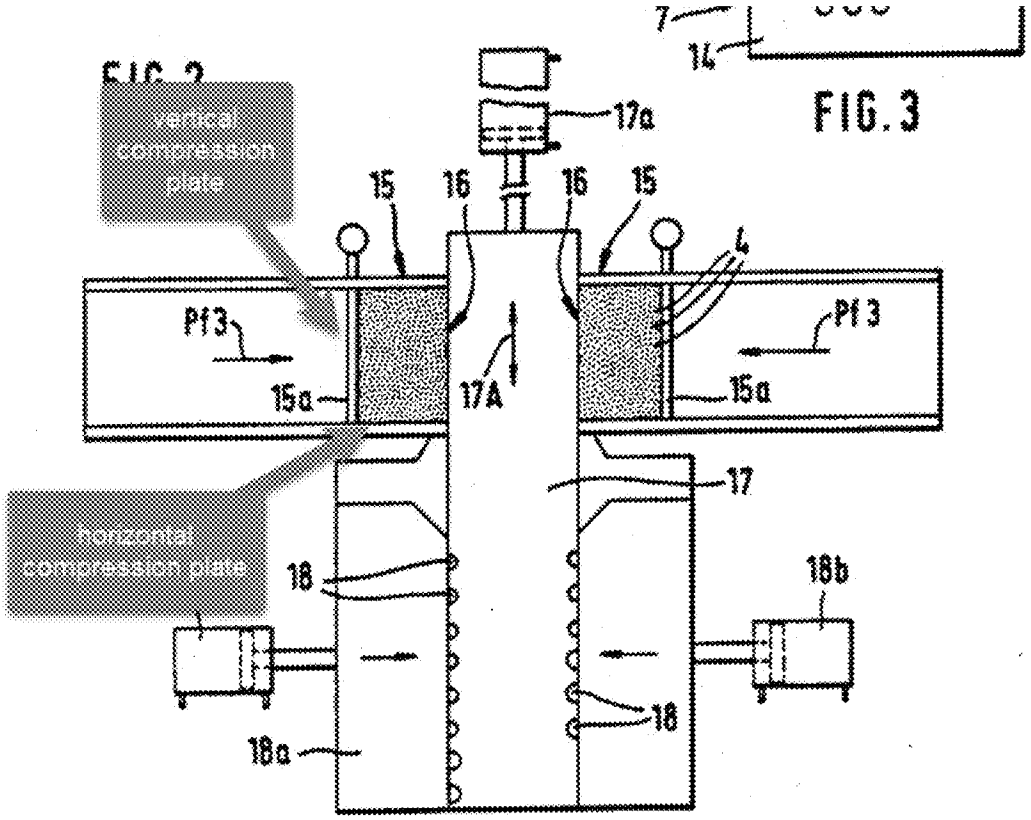
Regarding claim 12, and 14, Lorenz in view of Rueb in view of Morlock in view of Polese disclose the brush manufacturing machine according to claim 1, comprising the network of sensors and actuators controlled by the logic controller system that enables the pre-programmed sequence of processing steps (Morlock, Paragraph 23-53 discusses various scenarios in which sensors can be used; and Lorenz Column 6 Lines 32-57) .

Claims 3, 5-8, and 15 are rejected under 35 U.S.C. 103 as being unpatentable over Lorenz (US 4,310,377) in view of Rueb (US 5033797) in view of Morlock (EP3603446a1) in view of Polese (US 6,216,845) in view of Durant (US 4774799).

Regarding claim 3, Lorenz in view of Rueb in view of Morlock in view of Polese disclose the brush manufacturing machine according to claim 1, wherein the bristle feed mechanism comprises a magazine for the bristle tufts, having a vertical compression plate and a horizontal compression plate (Annotated Figure 2 of Rueb). Lorenz fails to explicitly disclose controlled by a solenoid valve and reed switch.

Durant teaches a feed mechanism controlled by solenoid valve (table XIII) and a reed switch (Column 7 Lines15-37). It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Lorenz to include the reed switch of Durant to help determine if there are bristles in the feed mechanism. Durant solves the same problem of using standard electronic components to determine if there is product in

the feed mechanism. The use of various sensors is viewed as a manner to make a manual process automated, which has been held to be of routine by one skilled in the art (see MPEP 2144.04). Durant calls the reed switch "a standard cut off device", which establishes that reed switches are known in the art for their capability of weighing and creating a signal when the weight is below a certain level. That signal can be used by a controller to shut off various solenoid valves.



Regarding claim 5, Lorenz in view of Rueb in view of Morlock in view of Polese in view of Durant disclose the brush manufacturing machine according to claim 3 wherein the bristle plant feeder comprises a conically shaped nozzle (Rueb Item 19 or 34).

Regarding claim 6, Lorenz in view of Rueb in view of Morlock in view of Durant disclose the brush manufacturing machine according to claim 5, wherein the conically shaped nozzle embodies a first opening, and a second opening, where a first diameter of corresponding

to the first opening is larger than the a second diameter of corresponding to the second opening (best shown in Figure 14 of Rueb).

Regarding claim 7, Lorenz in view of Rueb in view of Morlock in view of Polese in view of Durant disclose the brush manufacturing machine according to claim 6, wherein the conically shaped nozzle receives a distal end of a transport line (Rueb Item 10) at the first opening (Rueb Item 34 or 19).

Regarding claim 8, Lorenz in view of Rueb in view of Morlock in view of Polese in view of Durant disclose the brush manufacturing machine according to claim 7, wherein the conically shaped nozzle receives the bristle tufts through the transport line through the first opening to the second opening via a blower and suction system (Rueb Column 11 Line 59-Coluumn 12 Line 59).

Regarding claim 15, Lorenz in view of Rueb in view of Morlock in view of Polese in view of Durant disclose the brush manufacturing machine according to claim 3, wherein the bristle tuft picker comprises a picker plate with a picker face with one or more through holes with dimensions consistent with the through holes of the bristle carrier openings, the shape of the through holes is semi-circular (Rueb Figure 2 Item 18).

Claim 9 is rejected under 35 U.S.C. 103 as being unpatentable over Lorenz (US 4,310,377) in view of Rueb (US 5033797) in view of Morlock (EP3603446a1) in view of Polese (US 6,216,845) in view of Brown (US 2004/0048063).

Regarding claim 9, Lorenz in view of Rueb in view of Morlock in view of Polese disclose the brush manufacturing machine according to claim 1, wherein the ultrasonic fusion station shaped to match the bristle carrier profile and dimensions (Lorenz Figure 1 Item 4, Rueb Item 27). Lorenz fails to explicitly disclose wherein the ultrasonic fusion station comprises an ultrasonic horn.

Brown teaches a brush manufacturing machine wherein the ultrasonic fusion station comprises an ultrasonic horn (Paragraph 46, Item 34). It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Lorenz with the ultrasonic horn of Brown. The ultrasonic horn of Brown is used in close proximity with an anvil to melt plastic to the optimal degree while controlling the flow of excess material (Brown paragraph 13, 46-47).

Claim 11 is rejected under 35 U.S.C. 103 as being unpatentable over Lorenz (US 4,310,377) in view of Rueb (US 5033797) in view of Morlock (EP3603446a1) in view of Polese (US 6,216,845) in view of Jacobsen (US 5,588,688).

Regarding claim 11, Lorenz in view of Rueb in view of Morlock in view of Polese disclose the brush manufacturing machine according to claim 1. Lorenz fails to explicitly disclose wherein the unloading station comprises a presence sensor and a pick-and-place gripper.

Jacobsen teaches a robot arm that is capable of moving different small payloads in any desirable scenario to help with automated processes (Column 1 Lines 17-57). Jacobsen's robot arm comprises a presence sensor (Item 25) and a pick-and-place gripper (Item 34). It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to include a robotic arm to Lorenz similar to that taught by Jacobsen. Such a modification would reduce the need for human interaction .

Response to Arguments

Applicant's arguments, filed 5/4/2026, with respect to the rejection(s) of claim(s) 1 and 13 under 103- Lorenz (US 4,310,377) in view of Rueb (US 5033797) in view of Morlock (EP3603446a1) have been fully considered and are persuasive. Therefore, the rejection has been withdrawn. However, upon further consideration, a new ground(s) of rejection is made in

view of Lorenz (US 4,310,377) in view of Rueb (US 5033797) in view of Morlock (EP3603446a1) in view of Polese (US 6,216,845).

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to TOM R RODGERS whose telephone number is (313)446-4849. The examiner can normally be reached Monday thru Friday 8AM-5PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, David Posigian can be reached at (313) 446-6546. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/TOM RODGERS/
Primary Examiner, Art Unit 3723